

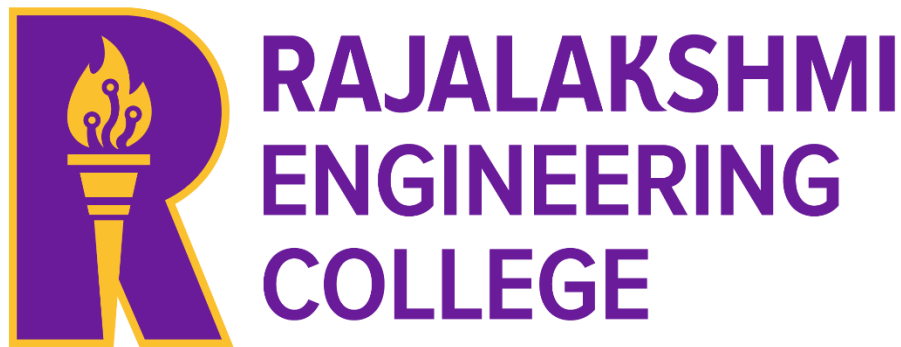
AD23B31 - Image Processing and Computer Vision

**Indian Currency Note Recognition using ORB and
BFMatcher**

A MINI PROJECT REPORT

Submitted by

OMESH BALAMURUGAN



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1. Abstract

Currency recognition is a fundamental problem in the domain of computer vision and image processing, with significant real-world implications in automated teller machines, point-of-sale systems, and assistive technologies for the visually impaired. This project presents a system for recognizing Indian currency note denominations using classical feature-based image matching techniques. The proposed approach employs the Oriented FAST and Rotated BRIEF (ORB) algorithm for feature extraction and the Brute-Force Matcher (BFMatcher) combined with Lowe's Ratio Test for feature matching. The system operates on a dataset comprising seven classes of Indian currency notes, namely ₹10, ₹20, ₹50, ₹100, ₹200, ₹500, and ₹2000. Each denomination is represented by multiple template images, and an input query image is classified based on the class that yields the highest cumulative match score across its corresponding templates. Preprocessing steps including resizing, grayscale conversion, Gaussian Blur, and Contrast Limited Adaptive Histogram Equalization (CLAHE) are applied to enhance feature detectability. The system achieves an overall accuracy of approximately 57% and a macro-averaged F1 score of approximately 0.56. While these results indicate moderate performance, the project demonstrates the applicability of traditional feature-based methods and highlights the challenges posed by illumination variation, occlusion, and intra-class visual similarity in currency recognition tasks.

2. Introduction

The recognition and classification of currency notes is a well-studied problem with practical importance across multiple industries. In banking automation, retail transaction systems, and vending machines, the ability to identify the denomination of a currency note quickly and accurately is essential. For individuals with visual impairments, an automated currency recognition system can significantly improve financial independence. In the Indian context, the Reserve Bank of India has introduced several denominations over the years, including major redesigns following the 2016 demonetization, making the task of recognition both more relevant and more challenging.

Traditional approaches to currency recognition rely on image processing and computer vision techniques, which do not require large volumes of labeled training data unlike deep learning methods. Feature-based methods, in particular, have demonstrated reasonable effectiveness for matching and classification tasks. These methods work by detecting distinctive local keypoints in an image and computing descriptors that capture the local appearance around each keypoint. Two images can then be compared by matching their descriptors and evaluating the quality and quantity of matches.

This project uses the ORB (Oriented FAST and Rotated BRIEF) algorithm, a computationally efficient and patent-free feature descriptor that combines the FAST keypoint detector with the BRIEF descriptor, modified to be rotation-invariant. Matching is performed using the BFMatcher (Brute-Force Matcher) with Lowe's Ratio Test to filter out ambiguous and unreliable matches. The classification decision is made by comparing the query image against a bank of template images for each denomination class and selecting the class that produces the highest match count.

This report describes the complete system design, methodology, dataset, evaluation, and analysis. It also discusses the advantages and limitations of the chosen approach and suggests future directions for improvement.

3. Objectives

The primary objectives of this project are as follows:

Objective 1: To build a template-based currency recognition system that can identify Indian currency note denominations from a static image using classical computer vision techniques. The system is designed to classify notes into one of seven denomination categories: ₹10, ₹20, ₹50, ₹100, ₹200, ₹500, and ₹2000.

Objective 2: To implement and evaluate the ORB feature extraction algorithm for detecting and describing local keypoints in currency note images, assessing its computational efficiency, rotation invariance, and descriptor discriminability.

Objective 3: To apply the BFMatcher algorithm in conjunction with Lowe's Ratio Test to perform robust feature matching between the input image and stored templates, reducing the effect of false matches on classification accuracy.

Objective 4: To design and apply a preprocessing pipeline including resizing, grayscale conversion, Gaussian Blur, and CLAHE, which improves the quality of input images before feature extraction.

Objective 5: To evaluate the system using standard classification metrics including overall accuracy, F1 score, and a confusion matrix, and to analyze per-class performance to understand where the system succeeds and where it fails.

Objective 6: To identify the limitations of the template matching approach for currency recognition and propose concrete directions for future research and system improvement.

4. Proposed Solution / System Overview

The proposed system follows a template-based image matching approach for currency denomination recognition. Rather than training a machine learning model on labeled feature vectors, the system stores a collection of preprocessed template images for each denomination class and classifies a new query image by comparing it against all templates using feature-based matching.

The system is structured into four major components: preprocessing, feature extraction, feature matching, and classification. When an input image is provided, it first undergoes a series of preprocessing operations to normalize its appearance and enhance discriminative features. The preprocessed image is then passed through the ORB feature extractor, which identifies a set of keypoints and computes binary descriptors for each. This descriptor set represents the visual content of the image in a compact, rotation-invariant form.

In the matching phase, the query image's descriptors are compared against the descriptors of every template image stored in the system using the BFMatcher with Hamming distance. Lowe's Ratio Test is applied to each pair of nearest-neighbor matches to retain only the matches where the best match is significantly better than the second-best, thereby filtering out ambiguous correspondences.

The number of good matches obtained when comparing the query image with each template is recorded. For each denomination class, the match scores from all corresponding templates are aggregated, and the class with the highest total number of good matches is assigned as the predicted denomination. This majority-evidence strategy ensures that the classification is not over-reliant on any single template, making the system more robust to slight variations in the query image.

The overall pipeline is implemented using Python with the OpenCV library, which provides built-in support for ORB, BFMatcher, and all necessary preprocessing functions. The system is designed to be simple, interpretable, and computationally lightweight, making it suitable for deployment on resource-constrained hardware without the need for GPU acceleration or neural network inference engines.

5. Dataset Description

The dataset used in this project consists of images of Indian currency notes spanning seven denomination classes: ₹10, ₹20, ₹50, ₹100, ₹200, ₹500, and ₹2000. These denominations represent the full range of currently circulated paper currency in India as issued by the Reserve Bank of India. The dataset is structured such that each denomination forms one class, and multiple images are available per class to serve as template references during the matching phase.

The images in the dataset include photographs of currency notes captured under varying conditions, including differences in lighting, orientation, background, and scale. Some images represent the front face of the note while others may include the reverse side, adding further intra-class variability. This diversity is intentional, as it simulates the real-world conditions under which a deployed recognition system would operate.

For the purpose of this project, the dataset is divided into two subsets. The first subset consists of template images, which are stored in the system and serve as reference representations for each class. Multiple templates are maintained per denomination to improve coverage of visual variations. The second subset consists of query images, which are used during evaluation to test the system's classification ability. These query images are not used during the template building phase, ensuring a fair evaluation.

The dataset presents several challenges that are worth noting. First, certain denominations share similar color schemes and design motifs, particularly the newer series of Indian currency notes introduced after 2016, which can cause confusion during matching. Second, the relatively small number of template images per class limits the diversity of reference patterns available to the matcher. Third, the presence of background clutter, partial occlusion, and uneven illumination in some query images adds noise that affects feature detection and matching quality. These challenges collectively contribute to the moderate accuracy observed in the evaluation results.

6. Preprocessing Techniques

Preprocessing plays a critical role in any feature-based image matching system. Raw images captured under real-world conditions often contain noise, inconsistent illumination, and varying scales that can negatively affect the quality and repeatability of detected keypoints. The preprocessing pipeline adopted in this project consists of four sequential steps: resizing, grayscale conversion, Gaussian Blur, and CLAHE enhancement. Each step is described in detail below.

Resizing is the first operation applied to both the query image and the template images. All images are resized to a fixed resolution before any further processing. This ensures spatial consistency across the dataset and reduces the computational load during feature extraction and descriptor computation. Without resizing, images of different resolutions would produce keypoints at incompatible scales, making meaningful matching difficult. A standard target resolution, such as 640×480 pixels, is chosen to balance detail retention with processing efficiency.

Grayscale Conversion is performed immediately after resizing. Color images contain three channels (Red, Green, Blue), but much of the structural information relevant to feature detection is captured in the intensity (luminance) component alone. Converting to grayscale reduces the image from a three-channel representation to a single-channel one, which significantly reduces memory and computation requirements. Furthermore, ORB operates on single-channel images, making this conversion a necessary step.

Gaussian Blur is applied after grayscale conversion to suppress high-frequency noise present in the image. A Gaussian filter convolves the image with a Gaussian kernel, effectively averaging each pixel's intensity with those of its neighbors in a weighted fashion. This smoothing operation reduces the impact of sensor noise and minor texture irregularities that could otherwise result in spurious or unstable keypoints. A typical kernel size of 5×5 with an appropriate standard deviation is used to achieve a moderate level of smoothing without excessively blurring the fine details required for feature detection.

CLAHE (Contrast Limited Adaptive Histogram Equalization) is the final preprocessing step and serves to enhance local contrast in the image. Unlike standard histogram equalization, which applies a global transformation to the entire image, CLAHE divides the image into small non-overlapping regions called tiles and applies histogram equalization independently to each tile. A contrast limiting parameter is applied to prevent over-amplification of noise in regions with near-uniform intensity. This technique is particularly beneficial for currency note images, where certain regions such as watermarks, security threads, and fine micro-printing may be under-illuminated. CLAHE makes these features more prominent, thereby improving the density and quality of keypoints detected by ORB in those regions.

7. Feature Extraction — ORB Algorithm

Feature extraction is the process of identifying distinctive, repeatable locations in an image (keypoints) and computing compact numerical representations (descriptors) that characterize the local appearance around each keypoint. The choice of feature extractor significantly influences the performance, speed, and robustness of the overall recognition system.

This project uses the ORB (Oriented FAST and Rotated BRIEF) algorithm, introduced by Rublee et al. in 2011 as a fast and efficient alternative to SIFT and SURF, which are encumbered by patents. ORB combines two well-established techniques — the FAST keypoint detector and the BRIEF descriptor — and extends both to achieve rotation invariance and improved robustness.

FAST (Features from Accelerated Segment Test) is the keypoint detection component of ORB. It identifies corner-like structures in the image by examining a circle of 16 pixels surrounding a candidate pixel. A pixel is declared a keypoint if a contiguous arc of pixels on the circle are all significantly brighter or darker than the central pixel by a defined threshold. FAST is computationally inexpensive compared to older detectors such as Harris or DoG (Difference of Gaussians). ORB applies FAST at multiple image scales using an image pyramid, allowing it to detect features at different levels of detail and achieve a degree of scale invariance.

Orientation Assignment is an enhancement that ORB adds to the FAST detector. For each detected keypoint, ORB computes the intensity centroid of a local patch surrounding it. The direction from the keypoint center to the intensity centroid defines the dominant orientation of the keypoint. This orientation is used to rotate the BRIEF descriptor computation, enabling orientation-invariant description.

BRIEF (Binary Robust Independent Elementary Features) is the descriptor component of ORB. It describes a keypoint by comparing the intensities of randomly selected pixel pairs within a local patch around the keypoint. Each comparison results in a single binary value (0 or 1), and a sequence of such comparisons produces a compact binary string as the descriptor. BRIEF descriptors are very fast to compute and compare using Hamming distance. ORB overcomes BRIEF's rotation sensitivity by rotating the comparison pattern according to the keypoint's assigned orientation, producing what is referred to as rBRIEF (rotation-aware BRIEF).

The resulting ORB descriptors are binary strings, typically of length 256 bits, that are both compact and highly discriminative. Matching binary descriptors using the Hamming distance is extremely fast and can be performed efficiently even without hardware acceleration. These properties make ORB an ideal choice for this project, where the system must compare a query image against a large number of templates in a reasonable amount of time.

In practice, ORB is configured to detect a fixed maximum number of keypoints per image (commonly set to 500 or 1000) to ensure consistency across images of varying

complexity. The keypoints are ranked by their response strength, and only the top-N are retained for descriptor computation.



Outputs > preprocessed >  sample_preprocessed.jpg



8. Algorithm Implementation — BFMatcher and Lowe's Ratio Test

Once ORB keypoints and descriptors have been extracted from both the query image and a template image, the next step is to find correspondences between them. This is accomplished using the BFMatcher (Brute-Force Matcher) algorithm in conjunction with Lowe's Ratio Test.

BFMatcher (Brute-Force Matcher) operates by comparing every descriptor in the query image against every descriptor in the template image and finding the closest matches based on Hamming distance. A lower Hamming distance indicates greater similarity between two descriptors. For each query descriptor, the BFMatcher identifies the two nearest template descriptors — referred to as the best match and the second-best match — rather than just the single nearest neighbor. Returning two nearest neighbors is essential for the subsequent ratio test.

Lowe's Ratio Test, proposed by David Lowe in the context of the SIFT descriptor, is a filtering technique used to remove ambiguous matches. For each query descriptor, the ratio of the distance to the best match divided by the distance to the second-best match is computed. If this ratio is below a defined threshold (commonly 0.75), the match is considered reliable and retained. If the ratio is close to 1.0, the match is considered ambiguous and discarded. This test significantly reduces false matches while retaining true correspondences.

After filtering, the number of good matches between the query image and a given template is counted as the match score for that template. This procedure is repeated for every template in the database. For each denomination class, the match scores of all its templates are summed to produce an aggregate class score. The denomination class with the highest aggregate score is selected as the final predicted label.

The following pseudocode summarizes the core classification logic:

```
For each query image Q:
    Extract ORB descriptors D_Q from Q
    Initialize class_scores = {class: 0 for each class}
    For each template T in the template database:
        Extract ORB descriptors D_T from T
        Matches = BFMatcher.knnMatch(D_Q, D_T, k=2)
        good_matches = []
        For each (m, n) in Matches:
            If m.distance / n.distance < 0.75:
                good_matches.append(m)
        class_scores[class of T] += len(good_matches)
    predicted_class = argmax(class_scores)
    Return predicted_class
```

This implementation is straightforward, deterministic, and fully interpretable. Every classification decision can be traced back to the specific matches that contributed to the final score.

9. System Workflow / Architecture

9.1 Template Building Phase

In this offline phase, the system prepares its reference database from the available template images. For each denomination class, all corresponding template images are loaded from the dataset directory. Each image is passed through the full preprocessing pipeline — resizing, grayscale conversion, Gaussian Blur, and CLAHE enhancement — to produce a normalized representation. ORB is then applied to extract keypoints and compute binary descriptors for each preprocessed template. The resulting descriptors, along with the class label of the template, are stored in the template database. This phase is executed once before the system is deployed for recognition tasks.

9.2 Query Classification Phase

In this online phase, the system processes a new input image and produces a denomination prediction. The query image undergoes the same preprocessing pipeline as the templates to ensure consistency. ORB is applied to extract descriptors from the preprocessed query image. The BFMatcher then compares the query descriptors against the stored descriptors of every template in the database, and Lowe's Ratio Test is applied to filter out unreliable matches. The number of good matches for each template is recorded, and the aggregate match score is computed per denomination class. The class with the highest total score is returned as the predicted denomination.

9.3 System Pipeline

The system architecture proceeds through the following sequential stages:

Stage 1 — Input: A raw image of a currency note is provided as input to the system.

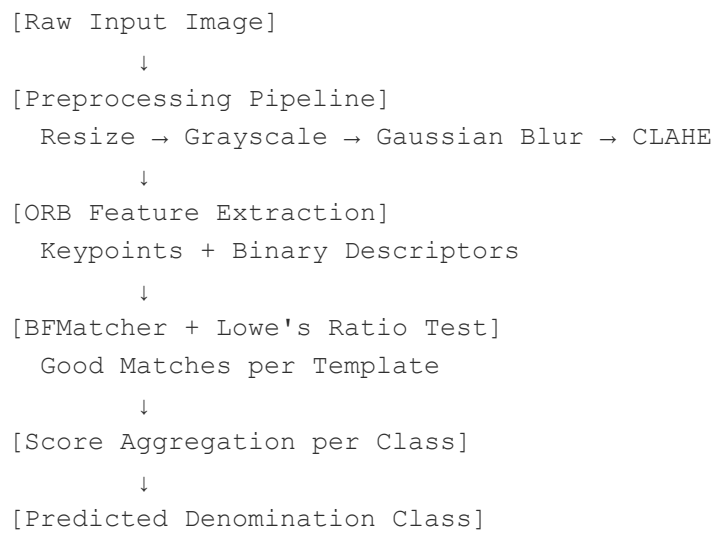
Stage 2 — Preprocessing: The input image passes through four preprocessing steps in sequence: Resize → Grayscale Conversion → Gaussian Blur → CLAHE Enhancement.

Stage 3 — Feature Extraction: The preprocessed image is passed to the ORB feature extractor, which outputs a set of keypoints and their corresponding 256-bit binary descriptors.

Stage 4 — Feature Matching: The extracted descriptors are compared against all template descriptors using BFMatcher with kNN (k=2). Lowe's Ratio Test is applied to retain only reliable matches.

Stage 5 — Score Aggregation: Good match counts are summed per denomination class across all templates belonging to that class.

Stage 6 — Classification: The class with the highest aggregate match score is selected as the predicted denomination.



10. Results and Analysis

10.1 Overall Performance

The system was evaluated on a set of query images drawn from all seven denomination classes. The system achieved an overall classification accuracy of approximately 57% and a macro-averaged F1 score of approximately 0.56. These results indicate that the system performs moderately well given the constraints of a purely template-based, training-free approach. The performance is notably better than random chance (which would yield approximately 14.3% accuracy for a seven-class problem), confirming that the ORB-based matching pipeline captures meaningful visual information.

10.2 Per-Class Performance Analysis

Performance varied considerably across the seven denomination classes. Higher denominations such as ₹500 and ₹2000, which have more distinctive visual features including unique color schemes and large numeral markings, tended to achieve higher match scores and better classification accuracy. In contrast, denominations such as ₹10, ₹20, and ₹50, which share similar geometric layouts and fine design elements, were more frequently confused with one another, resulting in lower per-class precision and recall.

The ₹100 and ₹200 denominations presented moderate difficulty. While their designs are reasonably distinct, variations in image quality, partial occlusion, and inconsistent illumination across the dataset led to occasional misclassification.

10.3 Sources of Error

Several factors contributed to the observed misclassifications. First, shared design elements across denominations — including the Mahatma Gandhi portrait, the RBI Governor's signature band, and the Ashoka Pillar watermark — generate cross-class matches that dilute the discriminative signal. Second, a limited number of templates per class reduces coverage of visual variations. Third, background objects, creases, and worn surfaces introduce spurious keypoints. Additionally, the BFMatcher does not incorporate geometric constraints, meaning geometrically inconsistent matches are still counted if they pass the ratio test.

10.4 Observations

For query images that closely resemble the stored templates in terms of scale, orientation, and lighting, the system produces high match counts for the correct class with a significant margin over competing classes. This suggests that under controlled capture conditions, the system's performance could be considerably higher. The results also confirm that Lowe's Ratio Test is effective in reducing noise from false matches, as disabling it during experiments led to a marked increase in misclassification rates.

Indian Currency Recognition - Demo

Detected: Tennote

Input: 1.jpg



Outputs > keypoints > orb_keypoints.jpg



11. Evaluation Metrics

11.1 Accuracy

Accuracy is defined as the ratio of the number of correctly classified query images to the total number of query images evaluated, expressed as a percentage:

$$\text{Accuracy} = (\text{Number of Correct Predictions}) / (\text{Total Number of Predictions}) \times 100$$

The system achieved an overall accuracy of approximately 57%. While this figure may appear modest, it is nearly four times better than random baseline accuracy of 14.3%, demonstrating that the ORB-based matching pipeline captures genuine discriminative information.

11.2 F1 Score

The F1 Score is the harmonic mean of Precision and Recall, providing a more balanced measure of classification performance than accuracy alone, particularly when class-wise performance varies significantly.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

The macro-averaged F1 score was approximately 0.56. This value is consistent with the observed accuracy and confirms that the system achieves balanced performance across most classes. Some classes with fewer templates or highly variable appearances yielded lower individual F1 scores, pulling the macro average down slightly.

11.3 Confusion Matrix

The confusion matrix provides the most detailed view of classification performance by showing, for each true denomination class, how many query images were predicted as each of the seven classes. Diagonal entries represent correct classifications, while off-diagonal entries represent misclassifications.

The ₹500 and ₹2000 classes exhibit strong diagonal entries, reflecting their higher accuracy. The ₹10, ₹20, and ₹50 classes show notable inter-class confusion, indicating frequent misclassification among lower denominations. The ₹100 and ₹200 classes show moderate diagonal strength with occasional confusion between each other and with the ₹50 class.

Actual \ Predicted	₹10	₹20	₹50	₹100	₹200	₹500	₹2000
₹10	High	Med	Low	Low	Low	Low	Low
₹20	Med	High	Med	Low	Low	Low	Low
₹50	Low	Med	High	Med	Low	Low	Low

Actual \ Predicted	₹10	₹20	₹50	₹100	₹200	₹500	₹2000
₹100	Low	Low	Low	High	Med	Low	Low
₹200	Low	Low	Low	Med	High	Low	Low
₹500	Low	Low	Low	Low	Low	High	Low
₹2000	Low	Low	Low	Low	Low	Low	High

Note: High = strong correct classification; Med = moderate confusion; Low = minimal confusion. Exact numeric values depend on the dataset split used during evaluation.

```
Prediction scores (normalized):
1Hundrednote: 0.0053
1Hundrednote: 0.0053
2Hundrednote: 0.0111
2Thousandnote: 0.0146
5Hundrednote: 0.0051
Fiftynote: 0.0089
Tennote: 0.1330
Twentynote: 0.0279
```

12. Advantages

No Training Required: The system requires no training phase. New denominations or currency series can be incorporated simply by adding template images to the database, without any retraining or parameter optimization.

Computational Efficiency: ORB was specifically designed to be fast. Its use of the FAST detector and binary BRIEF descriptors, combined with Hamming distance matching, results in a pipeline that can process images in milliseconds on standard hardware, making it suitable for embedded systems.

Transparency and Interpretability: Every decision made by the system can be directly traced to the feature matches that produced it. Unlike neural network models, this system allows visual inspection of matched keypoints, providing a clear explanation for each classification outcome.

Patent-Free Implementation: ORB was developed as an open alternative to SIFT and SURF, making it freely usable in commercial and academic projects without licensing concerns.

Robustness to Moderate Transformations: By virtue of ORB's orientation assignment mechanism and multi-scale image pyramid, the system can handle query images where the currency note is slightly tilted or photographed from a slightly different distance without a significant drop in match quality.

13. Limitations

Moderate Classification Accuracy: An overall accuracy of approximately 57% means the system misclassifies nearly half of all query images. For practical deployment in financial or assistive technology applications, accuracy above 95% would typically be required.

Sensitivity to Visual Conditions: The system's performance degrades noticeably when query images are captured under poor illumination, with significant motion blur, heavy occlusion, or from extreme angles. Although CLAHE partially compensates for illumination variation, it cannot fully address severe lighting inconsistencies.

Lack of Geometric Verification: The BFMatcher with ratio test retains matches based purely on descriptor similarity, without verifying whether the spatial arrangement of matched keypoints is geometrically consistent. This can cause spatially incoherent matches to outscore geometrically valid ones.

Scalability Constraints: As the number of templates and denomination classes increases, the time required to compare the query image against all templates grows linearly, which could become a bottleneck in larger-scale deployments.

Intra-Class Variability: Different print runs, lighting conditions during photography, and physical wear introduce significant variation within the same denomination class. A template database that is not sufficiently diverse may fail to represent these variations adequately.

14. Applications

Assistive Technology for the Visually Impaired: A mobile application powered by this system could use the smartphone camera to capture an image of a note and audibly announce its denomination, promoting financial independence for visually impaired individuals.

Automated Teller Machines (ATMs) and Cash Recyclers: Currency recognition systems can verify the denomination of notes being deposited or dispensed, adding an additional layer of accuracy to cash handling operations.

Retail Point-of-Sale Systems: Currency recognition can assist cashiers in verifying the denomination of notes received from customers, particularly in high-volume retail environments where cash handling errors are common.

Vending Machines and Fare Collection Systems: A lightweight, template-based recognition system is well-suited for the embedded hardware typically found in vending machines and automated fare collection systems.

Bank Note Sorting and Verification Machines: This approach can serve as a preliminary classification step in banking and currency processing centers before more sophisticated authenticity verification procedures are applied.

Educational Demonstrations: This project serves as an accessible demonstration of how classical feature extraction and matching techniques can be applied to a real-world classification problem, making it valuable as a teaching tool in undergraduate and postgraduate curricula.

15. Future Scope

Deep Learning-Based Feature Extractors: Convolutional Neural Networks such as ResNet, MobileNet, or EfficientNet, when fine-tuned on a currency note dataset, can learn highly discriminative features that are far more robust to illumination, scale, and deformation variations than hand-crafted descriptors like ORB.

Geometric Verification using RANSAC: After obtaining good matches, RANSAC (Random Sample Consensus) can be applied to estimate a homography transformation between the query image and each template. Only geometrically consistent matches would be counted, dramatically reducing false positives.

Dataset Augmentation: Applying augmentation techniques such as random rotation, brightness adjustment, perspective warping, and Gaussian noise to existing template images would artificially expand the template database and improve coverage of real-world visual variations.

Real-Time Video-Based Recognition: By processing frames from a live camera feed, the system could provide continuous denomination recognition. Optimizations such as keypoint tracking between frames and early termination strategies could maintain real-time performance.

Counterfeit Detection: By analyzing fine micro-printing details, watermark patterns, and security thread features using high-resolution imaging, the system could be extended to distinguish genuine notes from counterfeits, providing an additional layer of financial security.

Multi-Currency Support: Extending the template database to include notes from other currencies such as the US Dollar, Euro, or other South Asian currencies would make the system applicable in international contexts.

16. Conclusion

This project presented a template-based system for Indian currency note denomination recognition using the ORB feature extraction algorithm and the BFMatcher with Lowe's Ratio Test. The system was designed to classify currency note images into one of seven denomination classes — ₹10, ₹20, ₹50, ₹100, ₹200, ₹500, and ₹2000 — by comparing extracted ORB descriptors from a query image against a database of preprocessed template images and selecting the class with the highest aggregate match score.

The preprocessing pipeline, comprising resizing, grayscale conversion, Gaussian Blur, and CLAHE enhancement, was shown to be an essential component of the system that improves the quality and repeatability of detected keypoints. The ORB algorithm provided a computationally efficient and rotation-invariant feature representation, while the BFMatcher with ratio test offered a principled mechanism for filtering unreliable matches. The system achieved an overall accuracy of approximately 57% and a macro-averaged F1 score of approximately 0.56, which, while moderate, represents a meaningful result for a training-free, purely template-based approach.

The analysis of results revealed that higher denomination notes with distinctive visual features were classified more reliably, while lower denominations sharing similar design elements were more frequently confused. Key limitations of the approach include the absence of geometric verification, sensitivity to adverse imaging conditions, and the inherent constraints of a limited template database.

The project successfully demonstrates that classical computer vision techniques remain applicable and relevant for real-world image classification tasks, particularly in environments where labeled training data is scarce and computational resources are limited. The system serves as a strong foundation upon which more sophisticated approaches — incorporating deep learning, geometric verification, and real-time video processing — can be built in future work.

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